Reproducing COPOD: Data Integrity, Scoring Protocols and Limits of Generalisation

Shuharto Jannat
Student ID: 60778431
Master of Data Science, Macquarie University
COMP8240 Project Final Report

Abstract

I investigated whether COPOD's published outlier-detection results could be reproduced and whether its strengths transferred to additional data. A 30-dataset, ten-trial reproduction obtained mean ROC AUC 0.8132 and average precision (AP) 0.5475, compared with published 0.8247 and 0.5649. Aggregate similarity concealed four ARFF feature-selection errors, an additional label error, and unresolved MAT discrepancies. Controlled reruns distinguished changes in evaluation labels from changes in predictors. Five additional datasets received exploratory assessments, followed by a full APS truck-failure study and a ten-seed synthetic extension. On APS, Isolation Forest achieved mean AP 0.5860, exceeding batch COPOD's 0.5317 and a modified fixed-reference scorer's 0.5342. In a controlled dependence-violation setting, fixed COPOD's AP remained near prevalence while covariance scoring recovered the signal. A supplementary two-session LLM audit agreed on eight substantive claims but exposed citation-locator defects. The results show why data semantics and scoring-reference access must be checked before interpreting benchmark averages or generalisation claims.

1 Source paper and project goals

Outlier detection ranks unusual observations without requiring their anomaly labels during fitting. Its applications include equipment diagnostics, fraud screening and network monitoring, where costly positive cases may be rare. An unusual measurement pattern, however, is not automatically a verified failure or malicious event. My project investigates *COPOD: Copula-Based Outlier Detection*, introduced by Li et al. (2020).[1]

COPOD estimates empirical left and right tails for each feature, converts small probabilities into large negative-log contributions, and combines skewness-adjusted evidence into a score. For reference size m, an inclusive left-tail estimate is

$$\widehat{F}_j(x) = \frac{1}{m}\sum_{r=1}^{m} \mathbf{1}(X_{rj} \leq x). \tag{1}$$

Reversing the inequality gives the right tail. In the inspected implementation, each feature contributes the larger of its skewness-adjusted evidence and the average left/right evidence; contributions are summed across features. Although motivated by copulas, this score does not explicitly fit a covariance matrix or conditional feature relationships. Retained contributions explain the calculation, not the physical cause of an anomaly.

The paper claims competitive detection, efficient computation, no detector-specific hyperparameter tuning and interpretable contributions. It compares COPOD with nine alternatives on 30 datasets, using random 60/40 splits and ten trials. Its principal tables report ROC AUC and AP: COPOD has the highest mean of each, 0.8247 and 0.5649. My project concentrates on predictive results and implementation semantics. Recorded timings and feature diagnostics do not constitute a separate speed benchmark or human evaluation of explanation quality.

I selected this novel project for its research quality, openness and testable mechanism. IEEE ICDM is rated A* in CORE2023 (Computing Research and Education Association of Australasia, 2023); the paper offers explicit claims and a broad comparative evaluation. Released code/data and an implementation in PyOD (Zhao et al., 2019) make its behaviour inspectable. Most importantly, the featurewise construction suggests a controlled question about unusual values versus unusual relation-

[1] Paper: https://doi.org/10.1109/ICDM50108.2020.00135; accessible preprint: https://arxiv.org/abs/2009.09463.

ships. My scripts, results and documented changes are public.[2]

The project asks: **RQ1**, how closely can the released workflow recover the paper's results, and which differences can be explained; **RQ2**, how does COPOD transfer to additional existing data under explicit preprocessing and scoring protocols; and **RQ3**, which constructed anomaly mechanisms expose strengths or limitations? A separate LLM exercise checks evidence support for selected claims rather than replacing numerical evaluation.

1.1 Metrics and information access

ROC AUC measures positive–negative ranking, with half-credit for ties. AP is the non-interpolated sum of precision weighted by increases in recall (scikit-learn developers, 2026). It depends on prevalence and case mix, so scores from differently sampled datasets are not directly comparable measures of difficulty. Prevalence is a useful no-signal AP reference; finite-sample AP need not equal it exactly. I evaluate continuous anomaly scores, not contamination-based classification thresholds.

The scoring reference is another important distinction. Both the inspected authors' scorer and PyOD batch COPOD use training features together with the test batch when calculating test scores. Test labels are excluded, but test features influence tails and skewness. I additionally implemented a *fixed-training-reference* modification: inclusive tails and skewness come only from training data, with probabilities floored at $1/m$. It is explicitly a project modification. Training-score agreement and invariance to tested query chunking were checked before its comparative use.

2 Original datasets

The original evaluation uses 14 MAT variants from ODDS (Rayana, 2016) and 16 ARFF variants from the collection associated with Campos et al. (2016).[3] Table 1 describes every variant, including dimensions, positive fractions and domains. Many are repurposed classification datasets with designated outlier classes; their benchmark labels should not be interpreted as independently verified real-world anomalies. Repeated names identify differently processed files: for example, MAT WBC has 30 predictors while ARFF WBC has nine.

MAT files contain a numerical matrix `X` and binary label array `y`. ARFF files declare named attributes and data rows; names distinguish identifiers, measurements and the outlier label. Table 2 shows genuine partial examples. I preserved these released versions and audited their identities rather than replacing them with newer datasets. The large Mammography and Shuttle MAT inputs were subsampled by the historical workflow; their full source sizes and used sizes are distinguished in Table 1.

3 Replication and discrepancy investigation

3.1 Environment and execution

I ran the work in Linux GitHub Codespaces using Python 3.12.1. The recorded environment pins 25 packages, including NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.0, scikit-learn 1.9.0 and PyOD 3.6.4. These are the project's versions; the publication's exact historical environment was not recovered. The repository preserves the original source as a submodule and keeps project scripts separately.

Compatibility work reconciled the released `COD` import with its `COPOD` class, corrected file paths and seven ARFF names, converted measurements to numerical arrays, substituted the paper's Shuttle MAT entry for unavailable Cover, and used Feature Bagging instead of the driver's PCA to match the paper's comparison. These disclosed changes made execution possible; they are distinct from the later schema corrections.

All 30 datasets completed ten non-stratified 60/40 splits, seeds 0–9, with COPOD and ABOD, CBLOF, HBOS, Isolation Forest, KNN, LODA, LOF, One-Class SVM and Feature Bagging. The inspected scripts fit standardisation on training features and cap the two large MAT inputs at 10,000 rows. Labels evaluate scores and supply contamination settings, but are not passed to detector fitting. Each trial's metrics were rounded to four decimals before averaging, following the historical workflow. Repeated splits overlap and are not independent datasets.

Unit techniques and tools. Across the project, Python with NumPy, pandas and SciPy supported data loading, schema/missingness/duplicate audits and synthetic generation; PyOD and scikit-learn

[2] Project: `https://github.com/Shuharto-Jannat/copod-reproducibility-project`. Authors' source: `https://github.com/winstonll/COPOD`, pinned at `3ca1740`.

[3] Collections: `https://shebuti.com/about-odds/` and `https://www.dbs.ifi.lmu.de/research/outlier-evaluation/`. Exact files are supplied through the pinned authors' repository.

Format	Dataset	Used N	Used d	Positive %	Measurement domain
MAT	arrhythmia	452	274	14.60	ECG / cardiac rhythms
MAT	breastw	683	9	34.99	Breast cytology
MAT	cardio	1,831	21	9.61	Fetal heart measurements
MAT	ionosphere	351	33	35.90	Radar returns
MAT	lympho	148	18	4.05	Lymphography
MAT	mammography*	10,000	6	2.36	Breast screening
MAT	optdigits	5,216	64	2.88	Handwritten digit images
MAT	pima	768	8	34.90	Diabetes measurements
MAT	satellite	6,435	36	31.64	Satellite land cover
MAT	satimage-2	5,803	36	1.22	Satellite land cover
MAT	shuttle*	10,000	9	7.36	Spacecraft telemetry
MAT	speech	3,686	400	1.65	Speech features
MAT	wbc	378	30	5.56	Breast measurements
MAT	wine	129	13	7.75	Wine chemistry
ARFF	Arrhythmia	450	259	45.78	ECG / cardiac rhythms
ARFF	Cardiotocography	2,114	21	22.04	Fetal heart measurements
ARFF	HeartDisease	270	13	44.44	Cardiac measurements
ARFF	Hepatitis	80	19	16.25	Liver disease measurements
ARFF	InternetAds	1,966	1,555	18.72	Web advertisements
ARFF	Ionosphere	351	32	35.90	Radar returns
ARFF	KDDCup99	60,839	41	0.16 / 0.40†	Network connections
ARFF	Lymphography	148	18	4.05	Lymphography
ARFF	Pima	768	8	34.90	Diabetes measurements
ARFF	Shuttle	1,013	9	1.28	Spacecraft telemetry
ARFF	SpamBase	4,207	57	39.91	Email spam features
ARFF	Stamps	340	9	9.12	Document-stamp features
ARFF	Waveform	3,443	21	2.90	Synthetic waveforms
ARFF	WBC	223	9	4.48	Breast measurements
ARFF	WDBC	367	30	2.72	Breast diagnosis
ARFF	WPBC	198	33	23.74	Breast prognosis

Table 1: All 30 original benchmark variants. Counts describe the historical project inputs and label coding; d retains the released feature-selection semantics later audited. *Unseeded 10,000-row subsets of Mammography (11,183 rows) and Shuttle (49,097). †KDDCup99: 0.16% under the erroneous legacy label versus 0.40% under the named outlier label. Repeated names denote different processed versions, not interchangeable datasets.

Wine MAT: first two rows, selected columns				
Row	X_0	X_1	X_2	y
0	13.29	1.97	2.68	1
1	14.30	1.92	2.72	1

HeartDisease ARFF: selected attributes				
id	Age	Sex	Thal	outlier
1	.854167	1	0	yes
2	.791667	0	1	no

Table 2: Actual source excerpts, with most predictors omitted and Age rounded for display. Wine row numbers are locators, not inputs. ARFF source order is ID, measurements from Age through Thal, then outlier. Source identities were checked against the pinned files/audits.

Across 30 variants	ROC AUC	AP
Paper mean	.824700	.564920
Project mean	.813238	.547500
Project minus paper	−.011462	−.017420
Mean absolute dataset gap	.017683	.092157

Table 3: Historical COPOD reproduction with equal weight per dataset. The 60 paper cells match arXiv v1; Appendix A gives individual comparisons. Project values retain the released loader, including subsequently identified errors.

supplied detectors and ranking metrics; Matplotlib produced figures. Git and a pinned submodule separated upstream code from corrections, while committed plans, hashes, saved splits and scores supported provenance. Codespaces and a virtual environment supported execution; ACL LaTeX/BibTeX supported reporting. These choices demonstrate the assessed skills in data curation and analysis, data generation, computational project management and reproducible document preparation.

3.2 How closely were the results recovered?

The aggregate ROC AUC was relatively close, but AP's mean absolute dataset gap was much larger than its signed mean gap (Table 3). Positive and negative discrepancies therefore partly cancelled.

Method	Mean ROC AUC	Mean AP
COPOD	0.8132	0.5475
Isolation Forest	0.8055	0.5284
HBOS	0.7965	0.5086
One-Class SVM	0.7826	0.4907
CBLOF	0.7605	0.4655
KNN	0.7595	0.4466
LODA	0.7184	0.4334
LOF	0.6982	0.3787
ABOD	0.6890	0.3511
Feature Bagging	0.6869	0.3691

Table 4: Reproduced means across all 30 variants, ordered by ROC AUC. Each dataset has equal weight. These retain the historical loader and rounding; they are not corrected-schema rankings or statistical tests.

COPOD retained the highest unweighted mean ROC AUC and AP among the ten methods in the saved historical tables (Table 4). That finding describes the released-loader workflow, not a corrected benchmark ranking or a significance test.

Re-execution also has stochastic limits. Mammography and Shuttle were subsampled without a fixed seed before the seeded splits, and the sampled identities were not preserved. LODA lacked an explicit seed, while some baselines shared a mutable random state. The historical drivers suppressed warnings and applied `nan_to_num` to final scores. These facts limit exact replay and motivate focused audits instead of treating successful execution as sufficient validation.

3.3 Confirmed ARFF errors and controlled reruns

Auditing all 16 ARFF schemas found four positional feature-selection errors: an ID replaced Thal in HeartDisease, `dst_host_srv_rerror_rate` in KDDCup99, att19 in Lymphography and att8 in Pima. Unchanged predictor counts concealed these substitutions. The authors' and project loaders agreed, identifying shared released selection logic rather than a new discrepancy introduced by the audit.

KDDCup99 additionally used a measurement as the label. It produced 95 positives instead of 246 under the named outlier attribute; 339 rows disagreed. I separated this error from feature selection using three conditions: (A) legacy columns/labels; (B) the same scores with named benchmark labels; and (C) corrected columns with named benchmark labels. Four datasets, ten paired splits and three conditions produced 120 metric rows. All eight legacy mean checks recovered the historical project ROC/AP averages within 10^{-12} under the original rounding rule.

For KDDCup99, changing only labels moved AP from 0.260476 (A) to 0.449339 (B); correcting columns then reduced it to 0.343857 (C). The first change alters the evaluation target and prevalence while leaving scores identical. Table 5 shows why a lower metric can accompany a better-founded evaluation: correcting columns did not uniformly improve performance. Only COPOD received these controlled reruns, so they cannot establish a corrected ten-method ranking.

Numerical inspection found another hidden issue: all-zero KDD columns produced undefined intermediate contributions, which pandas row summation skipped to obtain finite scores. Thus finite final outputs did not certify every intermediate calculation. This observation is recorded separately from the label/feature corrections; I did not retrospectively rewrite the historical results.

3.4 Unresolved MAT and publication differences

Focused Wine, WBC and Satimage-2 reruns recovered all six project ROC/AP means within 10^{-12}. Full-precision AP means and sample SDs were 0.4324 ± 0.1520, 0.6953 ± 0.1061 and 0.7824 ± 0.0735. Their historical AP gaps from the paper remained -0.17580, -0.08741 and -0.07747.

Within each dataset, scores for every row were identical across the inspected splits, but test membership changed. Test positives ranged from 1–5 in Wine, 5–13 in WBC and 19–37 in Satimage-2. Metric variation therefore arose from the evaluated observations, including their identities and class composition. Score stability is consistent with the batch scorer referencing the same complete dataset in these runs. No non-finite contributions/scores or warnings were recorded in the 30 focused runs.

This establishes repeatability of the project averages and sensitivity to evaluation composition, but not the cause of the remaining publication gaps. Publication-era test membership and environment remain unknown. A separate post hoc hypothesis about misaligned AP table rows is retained in Appendix A; it is not used to replace the primary published comparison.

Dataset	ROC: B	ROC: C	AP: B	AP: C	Paired AP change ± SD
HeartDisease	.6505	.6979	.5967	.6444	$+.047704 \pm .013975$
KDDCup99	.9932	.9924	.4493	.3439	$-.105481 \pm .013851$
Lymphography	.9990	.9990	.9867	.9867	$.000000 \pm .000000$
Pima	.6439	.6492	.5261	.5223	$-.003888 \pm .015291$

Table 5: Feature correction with named benchmark labels fixed in both conditions: B uses legacy columns and C uses corrected columns. Means use full-precision trial metrics; SD describes ten paired split differences. ID removal and restoration of the omitted measurement are a joint intervention.

4 Construction of additional data

4.1 Five existing datasets and exploratory pilots

I assessed five publicly available datasets spanning financial applications, maintenance and cybersecurity. BAF is an existing third-party synthetic dataset; it is distinct from the data I constructed myself. Table 6 retains all pilot outcomes, including weak results. The pilots use one seed, 42, and PyOD batch COPOD. Their scripts and rounded results are preserved, but not the score-level bundles saved for later studies; I therefore treat them as exploratory assessments.

BAF Base. The source contains one million synthetic bank-account applications and 31 raw predictors (Jesus et al., 2022). I drew a stratified 20,000-row sample, encoded five categorical fields and retained documented -1 sentinels. The resulting 51-feature input retains a low-prevalence fraud setting.

RT-IoT2022. The released 123,117 network flows include benign device activity and attacks (Sharmila and Nagapadma, 2023). Removing an ID and duplicate feature-label records left 117,922 rows. I used all 12,015 unique benign flows and 1,335 sampled attacks, including all nine attack categories in the combined sample. This creates a 10% anomaly setting from an attack-dominated source; category coverage was not guaranteed separately in each split.

PhiUSIIL. This source contains 235,795 webpages represented by URL/content properties (Prasad and Chandra, 2024). I sampled 18,000 legitimate and 2,000 phishing rows, inverted the source labels so phishing is positive, and removed filename, URL, domain, TLD and title fields. Fifty numerical features remained, including specialised engineered properties whose ablation was not completed.

NATICUSdroid. The observed CSV has 29,332 rows with 86 binary permissions (Mathur, 2021). Inspection found 21,841 exact duplicate rows and 96 feature profiles with conflicting labels. Removing conflicting profiles and retaining one copy of each remaining profile gave 4,771 benign and 2,528 malware examples. All benign profiles and 530 sampled malware profiles formed the pilot. This label-informed cleaning defines a unique, unambiguous-profile population; it does not resolve conflicting ground truth for all applications.

All pilot scaling uses training statistics, but BAF and RT-IoT category vocabularies and constant-column screening use the full evaluation sample before splitting. These are preprocessing-information limitations. I selected APS for deeper study after the pilots because its supplied split, numerical measurements, missingness and rare failures supported focused questions. The selection was informed by outcomes, and part of its official test population had already been examined.

4.2 Full APS preparation

APS contains 60,000 training rows with 1,000 positives and 16,000 test rows with 375 positives, each with 170 anonymised numerical measurements (Scania CV AB, 2016).[4] Positive cases are APS-related component failures; negative cases are failures in other components, not necessarily healthy trucks. I retained the supplied partition and every row, including positive training cases.

The raw audit found 850,015 missing training cells and 228,680 missing test cells. Primary preprocessing dropped `cd_000`, which had only one distinct observed training value, leaving 169 predictors. A prespecified sensitivity additionally dropped eight predictors with strictly more than 50% training missingness, leaving 161. Training medians and standardisation parameters were ap-

[4]Dataset source: https://doi.org/10.24432/C51S51. Other additional dataset sources are linked through their cited records and the repository data-source guide.

Dataset	Train rows	Test rows	Used d	Test positives (%)	ROC AUC	AP
BAF Base	12,000	8,000	51	88 (1.10)	.5467	.0149
APS pilot	20,000	8,000	169	188 (2.35)	.9780	.5016
RT-IoT2022	8,010	5,340	92	534 (10.00)	.6891	.1360
PhiUSIIL	12,000	8,000	50	800 (10.00)	.9452	.7369
NATICUSdroid	3,180	2,121	86	212 (10.00)	.5796	.1072

Table 6: Recorded single-run pilots. Four non-APS datasets use stratified 60/40 splits. APS samples separately from the supplied training/test files. Approximately 10% positives in the last three datasets are experimental constructions, not natural prevalence; AP is not comparable as a cross-dataset league table.

plied to both partitions, without extra feature transformations or missingness indicators. Within each condition, methods shared the same prepared arrays; labels were excluded from preprocessing and fitting.

Column selection created one negative training duplicate pair in the primary condition and an additional negative train/test match in the secondary condition. Imputation/scaling introduced no further identical profiles, and there were no conflicting labels in these groups. All rows were retained as planned. Selected-feature equality neither identifies the same physical truck nor establishes independence of observations.

4.3 Synthetic mechanisms and controls

An initial study used five seeds, 42–46, and two Gaussian scenarios: a signed one-coordinate shift of magnitude four, and positive observations with broken feature dependence. The follow-up protocol was committed before its model runs and fixed seven settings with ten new data seeds, 100–109.

Each realization has 10,000 rows, 20 features and 5% designated positives. Normal observations form four independent blocks of five features, with within-block correlation ρ:

$$X_j = \sqrt{\rho}\, S_b + \sqrt{1-\rho}\, E_j, \qquad (2)$$

where S_b and E_j are independent standard normals before the block factor is shared. In the *shift family*, both groups begin with $\rho = 0.8$; each positive row receives a random signed shift in one uniformly selected coordinate, $\delta \in \{0, 2, 4, 6\}$. In the *dependence family*, normal $\rho \in \{0, 0.4, 0.8\}$ while positive features remain independent standard normal. The zero-strength setting in each family is a separate null control: labelled groups have the same population distribution.

The shift also alters relationships. Positive-feature variance is $1 + \delta^2/20$, reducing within-block correlation to $0.8/(1 + \delta^2/20)$. It is not a marginal-only intervention holding correlation fixed. Conversely, the dependence family matches population marginals while changing relationships when $\rho > 0$.

Within each family and seed, settings share base draws, row identities and splits to support paired contrasts; the 70 realizations are not independent datasets. Saved stratified 60/40 splits give 6,000 training rows including 300 positives and 4,000 test rows including 200 positives. No scaling, imputation or feature selection was applied. Existing datasets supplied their labels; constructed labels record generating origin, including the nulls. No new human annotation of observations was conducted, so annotation-agreement statistics do not apply.

Raw APS training data: three selected predictors

Row	class	aa_000	ab_000	ac_000
0	neg	76698	na	2130706438
9	pos	153204	0	182

Synthetic seed 100, $\delta = 4$: selected coordinates

Row	label	x11	x12	x13
0	0	$-.3777$	$-.3997$	$-.2599$
16	1	-2.1609	-2.2229	-6.2985

Table 7: Genuine partial new-data excerpts. APS tokens are raw; `na` denotes missingness. Synthetic values are rounded: row 16's x13 was -2.2985 before its -4 intervention. Row indices are locators, not features. Raw hashes and a complete hash-matching one-condition generator replay establish excerpt provenance.

5 Results on additional data

5.1 APS: comparator, reference and missingness

The full APS study compares batch COPOD, fixed-reference COPOD and Isolation Forest (Liu et al., 2008). Isolation Forest uses 200 trees, 256 sampled rows per tree, all features, no bootstrap, and seeds 42–46; scores are negative `score_samples`. No best seed was selected. Batch COPOD processes

the full 16,000-row test set together; the fixed-reference probability floor is $1/60{,}000$. Two conditions yield four deterministic COPOD results and ten Isolation Forest fits.

Isolation Forest exceeded both COPOD procedures on both metrics for every seed in each condition (Table 8). Its primary AP advantage over fixed COPOD was 0.051723. In contrast, fixing the COPOD reference changed primary AP by only 0.002490. Removing eight heavily missing features reduced AP by 0.002330 for batch COPOD, 0.002213 for fixed COPOD and 0.016087 for the Isolation Forest mean. Every paired seed also worsened. Missingness alone therefore did not justify this removal choice on the supplied split.

I quantified the small fixed-minus-batch difference using 2,000 paired class-stratified resamples of the saved primary scores. Each replicate draws 375 positives and 15,625 negatives with replacement, using identical indices for both methods. The AP difference's 95% percentile interval is [0.001395, 0.003893]; the ROC difference is 0.000131 with interval [0.000101, 0.000165]. These condition on fitted models, preprocessing, observed scores, prevalence and the original batch reference. They exclude retraining, rescoring when test-batch composition changes and future-data uncertainty; no practical-equivalence margin was specified.

As a retrospective top-rank diagnostic, precision at 375, with fractional allocation at cutoff ties, was 0.549333 for both COPOD procedures and 0.583467 ± 0.003477 for Isolation Forest. The cutoff uses the known positive count, so it is not a deployable alert rule. Recorded full-data warnings and non-finite final-score counts were zero; training-score agreement and tested fixed-scorer batch invariance passed. These checks support the implemented comparisons, not universal detector correctness.

5.2 Synthetic results: a mechanism-specific boundary

The synthetic extension adds squared Mahalanobis distance from an estimated mean and Ledoit–Wolf shrunk covariance (Ledoit and Wolf, 2004). It is a project scoring procedure built from that estimator, not an oracle using the generating covariance. I selected it to represent the generator's Gaussian linear relationships: its advantage tests signal recoverability in this design, not universal detector superiority. All three methods fit the same contaminated training rows without labels. The fixed COPOD floor is $1/6000$. Its comparison with covariance scoring holds training-only information access constant.

The initial five-seed study already showed a contrast: batch COPOD mean AP was 0.1024 for shifts and 0.0490 for dependence violations. The extension makes the comparison more informative by varying strength, adding nulls and adding a relationship-sensitive comparator (Figure 1). Both COPOD procedures' mean AP increased across tested shift strengths, reaching 0.1008 and 0.1004 at $\delta = 6$. This concerns group means, not monotonic improvement in every seed.

At dependence $\rho = 0.8$, batch and fixed COPOD AP were 0.0503 and 0.0504, near the 0.05 prevalence reference, while ROC AUC was about 0.55. Ledoit–Wolf scoring achieved ROC AUC 0.9967 and AP 0.9763. Its paired advantage over fixed COPOD was 0.445921 ± 0.018696 ROC AUC and 0.925907 ± 0.007094 AP, positive in all ten seeds. The signal was therefore recoverable by a covariance-sensitive score even though the tested COPOD procedures ranked positives poorly near the top. Correlated normal features usually move together; independent positives can violate low-variance relationship directions while their individual values remain ordinary. Mahalanobis scoring is sensitive to those directions.

Changing COPOD's reference produced small, inconsistently signed differences and did not resolve this dependence-setting weakness. Across both nulls and all methods, mean ROC AUC ranged 0.5015–0.5091 and AP 0.0512–0.0536. All 210 planned results were retained, with no outcome-based seed rejection; construction/numerical checks passed and recorded model warnings/non-finite scores were zero. SD measures variation across generated data, unlike APS's model-seed SD. The design fixes dimension, sample size and prevalence and does not establish performance on non-Gaussian, local or clustered anomalies.

5.3 Relationship to the reproduction

The additional studies qualify what the favourable benchmark average implies. The five pilots ranged from weak BAF/permission-profile discrimination to stronger APS and phishing rankings; the full APS comparator then outperformed COPOD, while the synthetic study exposed a specific boundary hidden by dataset averages. This qualifies a broad interpretation of competitiveness without disproving the paper's reported benchmark result.

Condition	Method	ROC AUC	Average precision
Primary (169 predictors)	Batch COPOD	.977690	.531741
	Fixed-reference COPOD	.977821	.534231
	Isolation Forest	.985060 ± .000313	.585954 ± .011787
Drop high missingness (161)	Batch COPOD	.976845	.529411
	Fixed-reference COPOD	.976990	.532017
	Isolation Forest	.984477 ± .000418	.569867 ± .008575

Table 8: Full supplied APS test partition, prevalence 2.34375%. Isolation Forest entries are means ± sample SD across five model seeds. COPOD has one deterministic result per condition. Fixed COPOD and Isolation Forest use training-only references; batch COPOD additionally uses test features.

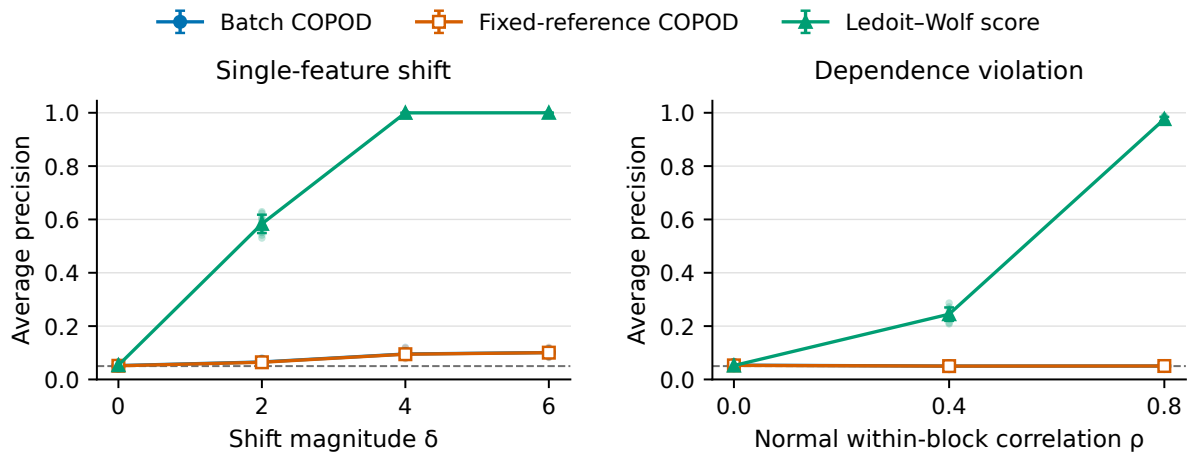


Figure 1: Synthetic AP across ten data seeds. Large markers/lines show means, bars show sample SD and faint dots show individual seeds. Dashed lines mark 5% prevalence. Left: signed single-feature shifts with normal correlation 0.8. Right: normal correlation increases while positive features stay independent. Zero strength defines a separate null within each family. Batch and fixed COPOD curves nearly overlap.

Evaluation also changed. The reproduction uses the authors' scorer, ten overlapping non-stratified splits and heterogeneous benchmark labels. Pilots use one mostly stratified construction; APS uses its supplied partition and five comparator seeds; synthetic results vary data generation over fixed-size stratified partitions. New studies use inspected PyOD plus the explicitly modified fixed scorer, different preprocessing, and different class proportions. ROC/AP remain common definitions, but neither higher APS scores nor lower synthetic scores measure a controlled improvement or deterioration relative to the 30-dataset mean.

6 LLM-as-judge evaluation

Deterministic metrics are primary because scores and labels permit direct calculation. In the judging exercise, I used an LLM to check whether selected report statements matched frozen evidence. The protocol supplied six evidence blocks with 38 excerpts, eight substantive claims and two diagnostic claims. Two fresh conversations used the configured ChatGPT model choice, labelled GPT-5.6 Sol in the menu and 5.6 Pro in the composer; Session B reversed the claim order. Exact backend versions, temperature and seed were not exposed; identical thinking effort was not verified. Known judge biases motivate this bounded role (Zheng et al., 2023).

Observed outcome	A	B
Substantive agreement with reference	8/8	8/8
Diagnostics match expectations	2/2	2/2
Citation entries	29	31
Locator defects found	2	0

Table 9: The reference was AI-drafted and author-approved; citation review was AI-assisted. A/B substantive agreement was also 8/8. These are descriptive agreement counts, not a general accuracy estimate.

Both sessions supported the eight substantive claims. They contradicted the diagnostic assertion that fixed COPOD's dependence-setting AP was about 0.5000 rather than 0.0504, and found insuf-

ficient evidence that synthetic results established superiority on future APS observations. Diagnostic expectations were withheld from the prompts. Session A nevertheless cited the wrong row twice for the first diagnostic: E05.01 row 21 instead of row 20, despite quoting the correct figures. Verdict agreement did not guarantee accurate citations.

The initial reference and my later acceptance of all ten verdicts were guided by AI assistance, not independent human validation. All substantive claims were supported, limiting category diversity; two sessions cannot distinguish an order effect from generation variability. Visible settings and fresh chats also do not prove complete isolation. The archive contains one supplied complete response per session; first-generated status and exact execution timestamps were not independently established. I preserved raw responses, prompt identities, audits and guided decisions rather than treating the LLM as experimental ground truth.

7 Ethical considerations

The Australian Code requires honesty, rigour, transparency, respect and accountability (National Health and Medical Research Council et al., 2018). I apply the unit readings as a framework: Benotti and Blackburn (2022) ask about benefits, harms and vulnerable groups; Karamolegkou et al. (2025) distinguish concerns, actions taken and further safeguards. These NLP studies guide reflection rather than establish the safety of this anomaly-detection project.

Benefits, harms and affected people. Reproducible code and corrected data semantics can help researchers avoid misleading comparisons. A suitably validated detector could help fleet operators prioritise inspection or security teams investigate suspicious events. However, missed APS failures could affect operators and road users; false alerts impose inspection costs and downtime. Fraud false positives could obstruct legitimate banking access; security false positives could wrongly restrict users or software. People with fewer resources to contest errors, or whose behaviour is poorly represented, could bear disproportionate costs. These are plausible downstream risks, not measured outcomes here. No demographic fairness audit or affected-stakeholder consultation was conducted. Deployment would require representative validation, subgroup error checks, cost-sensitive thresholds and accountable review of consequential decisions. ROC/AP and the retrospective top-375 cutoff do not establish those safeguards.

Data, representation and privacy. Original medical benchmarks and new permission/network datasets can reflect sensitive behaviour despite omitted identifiers. Classification-derived outliers do not establish clinical abnormality or wrongdoing. BAF is synthetic but models a consequential financial setting; the Gaussian data carry construction labels, not observed failures. I collected no new personal records and attempted no re-identification. Label-informed conflict removal and constructed class ratios change the evaluated population, so I report those decisions and avoid describing sampled prevalence as operational prevalence. This does not resolve representation bias or establish that original collection and consent conditions were adequate.

Access and attribution. Public access is not a blanket redistribution licence. Current UCI records list CC BY 4.0 for RT-IoT2022, PhiUSIIL and NATICUSdroid (Sharmila and Nagapadma, 2023; Prasad and Chandra, 2024; Mathur, 2021). APS combines a CC BY banner with original distribution text naming GPL v3-or-later (Scania CV AB, 2016). BAF's publisher datasheet specifies CC BY-NC-ND 4.0, unlike the earlier project note's SA description (Jesus et al., n.d.). I preserve these discrepancies, credit sources and keep downloaded additional raw datasets outside Git. Hashes and the accompanying acquisition guide support traceability; they do not resolve licence conflicts or grant permission to redistribute modified records. Provider clarification remains necessary where terms are unclear.

Publication, resources and residual risk. Reporting weaknesses supports defensive evaluation, but public methods could also help adversaries study detection limits. I present controlled failure modes without claiming operational robustness. Aggregate rankings, synthetic success and selected APS results are bounded to their tested settings. Preserved outputs, separate corrections and unresolved findings support the Code's responsibilities R21–R23 and R26–R27; loader errors do not establish author misconduct. Saved scores are reused for summaries, resampling and figures, avoiding unnecessary detector refits. Total energy use and emissions, including hosted LLM use, were not

measured, so computational efficiency is not presented as demonstrated environmental benefit.

AI assistance and responsibility. Generative AI assisted planning, code drafting/debugging, evidence checks, interpretation and report preparation. I executed the recorded workflow and accepted guided review decisions; I remain responsible for the submission. The AI-drafted reference and my approval are disclosed, not presented as independent human validation. Raw judge responses and citation defects are preserved because fluent explanations and agreement can encourage unjustified trust. The audit checks bounded statements; it neither certifies the experiments nor replaces source-based review.

8 Conclusions and reflection

RQ1 has a qualified answer: I completed the released-workflow comparison and recovered project results in focused reruns, but aggregate agreement concealed semantic errors and unresolved publication gaps. The ARFF investigation shows why correct measurement and label definitions matter more than obtaining a higher score. MAT diagnostics distinguish reproducibility of project averages from explanation of the original paper's numbers.

For RQ2, all five pilots remain informative, including weak outcomes. Full APS supports the specified Isolation Forest comparator on the supplied split, shows a small COPOD reference effect, and finds no benefit from the tested high-missingness removal. For RQ3, covariance scoring recovered the deliberately constructed relationship anomalies much better than COPOD, while stronger single-feature shifts improved COPOD mean AP. These are bounded conclusions about the tested populations and mechanisms.

The project became more focused than its proposal. Broad factorial variation in dimensions, prevalence and marginal families, local/cluster anomalies, additional pilot prevalence sweeps and phishing-feature ablations were not completed. I instead developed a traceable APS case study and paired synthetic extension. The LLM plan also narrowed from broad paper-claim judging to support checks on fixed statements, with an explicitly assisted reference. This scope change made data semantics and information access the central contribution rather than a catalogue of additional scores.

My main lesson is to investigate what a metric measures before trying to close its gap. Preserved row identities, predictions and controlled comparisons made limitations more interpretable. Priority future work is to freeze remaining stochastic sampling, verify publication-era inputs/tables, rerun all baselines under corrected schemas, and broaden the synthetic and deployment-relevant validation. The present evidence establishes what was recovered and corrected without pretending that the remaining uncertainty has disappeared.

References

Luciana Benotti and Patrick Blackburn. 2022. Ethics consideration sections in natural language processing papers. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4509–4516, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Guilherme O. Campos, Arthur Zimek, Jörg Sander, Ricardo J. G. B. Campello, Barbora Micenková, Erich Schubert, Ira Assent, and Michael E. Houle. 2016. On the evaluation of unsupervised outlier detection: Measures, datasets, and an empirical study. *Data Mining and Knowledge Discovery*, 30(4):891–927.

Computing Research and Education Association of Australasia. 2023. CORE2023 conference rankings: IEEE international conference on data mining. Historical CORE2023 entry; accessed 31 October 2026.

Sérgio Jesus, José Pombal, Duarte Alves, André Cruz, Pedro Saleiro, Rita Ribeiro, João Gama, and Pedro Bizarro. 2022. Turning the tables: Biased, imbalanced, dynamic tabular datasets for ML evaluation. In *Advances in Neural Information Processing Systems*, volume 35, pages 33563–33575. Curran Associates, Inc.

Sérgio Jesus, José Pombal, Duarte Alves, André F. Cruz, Pedro Saleiro, Rita P. Ribeiro, João Gama, and Pedro Bizarro. n.d. BAF dataset suite datasheet. Feedzai. Licence statements in question 44 and Author Statement; accessed 31 October 2026.

Antonia Karamolegkou, Sandrine Schiller Hansen, Ariadni Christopoulou, Filippos Stamatiou, Anne Lauscher, and Anders Søgaard. 2025. Ethical concern identification in NLP: A corpus of ACL anthology ethics statements. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11618–11635, Albuquerque, New Mexico. Association for Computational Linguistics.

Olivier Ledoit and Michael Wolf. 2004. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.

Zheng Li, Yue Zhao, Nicola Botta, Cezar Ionescu, and Xiyang Hu. 2020. COPOD: Copula-based outlier detection. In *2020 IEEE International Conference on Data Mining*, pages 1118–1123. IEEE.

Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. 2008. Isolation forest. In *2008 Eighth IEEE International Conference on Data Mining*, pages 413–422. IEEE.

Akshay Mathur. 2021. NATICUSdroid (Android permissions). UCI Machine Learning Repository. Dataset.

National Health and Medical Research Council, Australian Research Council, and Universities Australia. 2018. Australian code for the responsible conduct of research. Commonwealth of Australia, Canberra.

Arvind Prasad and Shalini Chandra. 2024. PhiUSIIL phishing URL (website). UCI Machine Learning Repository. Dataset; the repository cites the accompanying article DOI 10.1016/j.cose.2023.103545.

Shebuti Rayana. 2016. ODDS library. Stony Brook University, Department of Computer Science.

Scania CV AB. 2016. APS failure at scania trucks. UCI Machine Learning Repository. Dataset.

scikit-learn developers. 2026. average_precision_score. scikit-learn documentation. Accessed 31 October 2026.

B. S. Sharmila and Rohini Nagapadma. 2023. RT-IoT2022. UCI Machine Learning Repository. Dataset.

Yue Zhao, Zain Nasrullah, and Zheng Li. 2019. PyOD: A python toolbox for scalable outlier detection. *Journal of Machine Learning Research*, 20(96):1–7.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623. Curran Associates, Inc.

A Complete historical comparison and unresolved diagnostics

Table 10 gives all saved COPOD comparisons, with paper cells checked against arXiv v1. Means preserve the historical per-trial rounding and released loader. The four affected ARFF rows are not replaced by later corrected results. The final IEEE table and publication-era data identities were not independently recovered.

Format	Dataset	Paper ROC	Project ROC	Paper AP	Project AP
MAT	arrhythmia	0.8021	0.8134	0.4727	0.4897
MAT	breastw	0.9936	0.9935	0.9877	0.9877
MAT	cardio	0.8974	0.9214	0.5793	0.5902
MAT	ionosphere	0.8307	0.7992	0.7194	0.6808
MAT	lympho	0.9935	0.9948	0.8936	0.9053
MAT	mammography	0.8942	0.9128	0.4290	0.4318
MAT	optdigits	0.7328	0.6798	0.0532	0.0440
MAT	pima	0.6638	0.6492	0.5407	0.5222
MAT	satellite	0.6612	0.6328	0.5854	0.5697
MAT	satimage-2	0.9852	0.9784	0.8599	0.7824
MAT	shuttle	0.9980	0.9947	0.9807	0.9585
MAT	speech	0.4845	0.4728	0.0195	0.0220
MAT	wbc	0.9747	0.9590	0.7827	0.6953
MAT	wine	0.9490	0.8909	0.6082	0.4324
ARFF	Arrhythmia	0.7618	0.7663	0.7519	0.7582
ARFF	Cardiotocography	0.6370	0.6642	0.3778	0.4009
ARFF	HeartDisease	0.6728	0.6505	0.6400	0.5967
ARFF	Hepatitis	0.8432	0.8446	0.5849	0.5650
ARFF	InternetAds	0.6763	0.6768	0.5102	0.5112
ARFF	Ionosphere	0.8219	0.7894	0.5302	0.6620
ARFF	KDDCup99	0.9972	0.9976	0.5667	0.2605
ARFF	Lymphography	0.9985	0.9990	0.4637	0.9867
ARFF	Pima	0.6466	0.6439	0.7031	0.5261
ARFF	Shuttle	0.8795	0.8330	0.2389	0.0786
ARFF	SpamBase	0.7007	0.6761	0.9817	0.5283
ARFF	Stamps	0.9388	0.9424	0.0854	0.4763
ARFF	Waveform	0.7865	0.7234	0.0785	0.0600
ARFF	WBC	0.9904	0.9904	0.8381	0.8381
ARFF	WDBC	0.9826	0.9730	0.8407	0.8265
ARFF	WPBC	0.5465	0.5337	0.2438	0.2378

Table 10: All 30 historical comparisons, rounded to four decimals for display. Format distinguishes similarly named dataset versions. The complete ten-method tables and recorded runtimes remain in the public repository.

A post hoc AP-alignment audit used five baseline methods, excluding COPOD from its matching criterion, to propose a seven-row reassignment in the printed ARFF AP table. COPOD ARFF AP mean absolute error would fall from 0.142178 to 0.016361. This remains a hypothesis, not a confirmed erratum; the as-recorded comparison is primary. In the KDD numerical audit, each affected scoring run had 121,678 undefined intermediate cells from two all-zero columns over 60,839 rows. Twenty runs were flagged, while final row sums stayed finite. Neither observation was used to rewrite the historical benchmark.

B Additional performance views

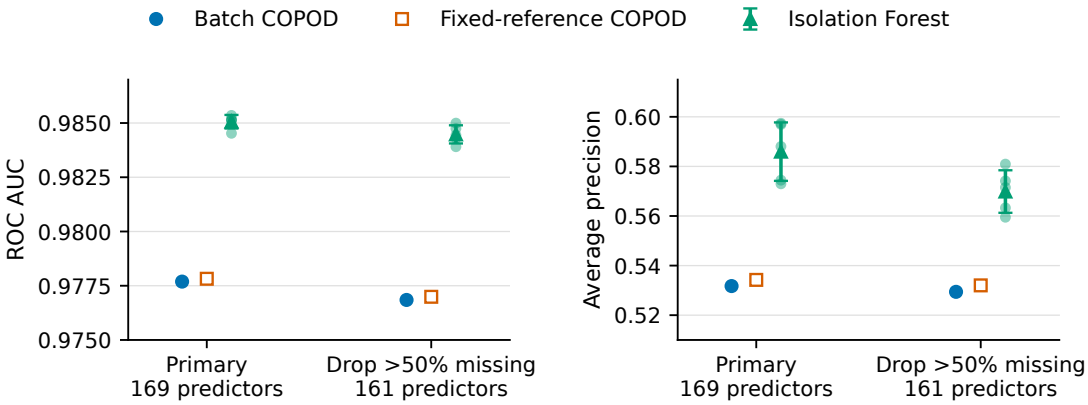


Figure 2: APS seed-level view of Table 8. Dots show saved results and bars the sample SD of the five Isolation Forest seeds; axes are zoomed. Deterministic COPOD has no model-seed SD. These are not test-population confidence intervals.

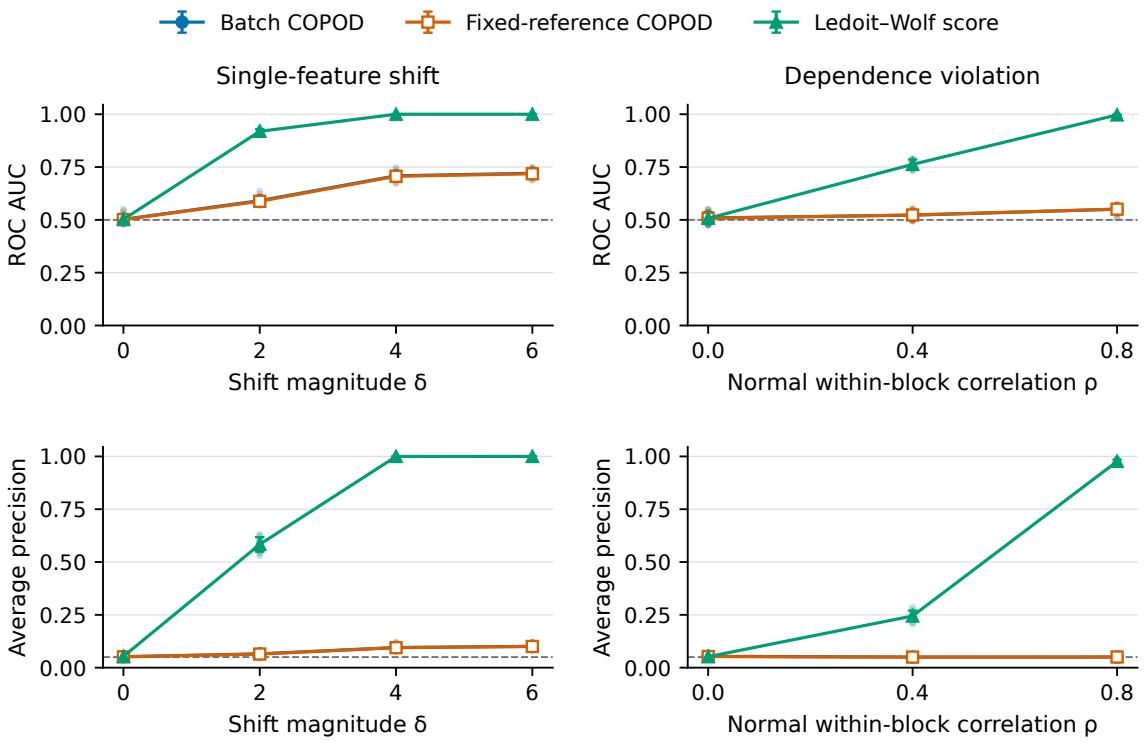


Figure 3: Synthetic ROC AUC and AP. Means, sample SD and all seed values are shown. Dashed references are ROC AUC 0.5 and AP prevalence 0.05. Dependence-setting COPOD ROC around 0.55 coexists with AP near prevalence; the metrics answer different ranking questions.

C Reproducibility and evidence locations

The supplied archive records project commit `3ab30e5`; the report evidence manifest preserves its full identifier and archive hash. The original authors' submodule is pinned at `3ca1740`. A full Git clone with submodules and history is required to reproduce workflows that inspect source files and historical commits; a source ZIP alone omits the submodule contents. Recorded dependencies are in `environment/requirements.txt`. The accompanying reproducibility and data-acquisition guides distinguish setup, regeneration, verification of frozen results and new model runs. They are report-package additions for a subsequent repository update, not files already present at this evidence checkpoint.

Reproduction and corrections

`results/comparison/`, the two `results/reproduction_*_10_trials/` directories, `results/audits/arff_dataset_mapping/`, `results/loader_corrections/v1/` and `results/audits/mat_reproduction_gaps/v1/` retain historical means, schema findings, controlled reruns and remaining gaps.

Five pilots

The five `scripts/smoke_test_*.py` files for additional data and `notes/new_data_feasibility.md` preserve construction and rounded results. Original-benchmark smoke-test result folders are not evidence bundles for these five pilots.

APS

`results/audits/aps_full_data/v1/` and `results/aps_full_evaluation/v1/` contain integrity checks, 14 results, score vectors, summaries and 2,000 paired resampling replicates. Raw provider files must be obtained separately.

Synthetic data

Initial and extension generators define the raw measurements, which are Git-ignored. The extension's defaults cover seed 100; the full study explicitly requests seeds 100–109. Saved configuration, splits, hashes, 210 result rows and paired summaries are in `results/synthetic_extension/v1/`.

LLM audit

`results/llm_judge/v1/` retains frozen/submitted prompts, raw/parsed responses, citation reviews, summary tables and later final guided author decisions. Its summary script sends no LLM requests. Earlier pending-review metadata remains a historical snapshot, supplemented by the later adjudication.

During report preparation, saved APS and synthetic metrics were independently recalculated from score vectors, paired summaries/intervals checked, LLM summary outputs verified, raw original/APS snippets hash-checked, and one synthetic generator condition replayed for its excerpt. The report figures were redrawn from unchanged saved metrics. These editorial checks neither refitted detectors nor produced new judge responses. Command paths/options were checked, but a fresh clean installation and full end-to-end rerun were not performed. Unseeded historical sampling and randomness remain exact-replay limitations.